AI-Based Waste Segregation: A CNN and DNN Approach

AJAY CHRISTO P

UG Scholar
Artificial Intelligence and Data Science
Karpagam Institute of Technology
Coimbatore, Tamil Nadu, India
christoajay007@gmail.com

DHARANIDHARAN S

UG Scholar
Artificial Intelligence and Data Science
Karpagam Institute of Technology
Coimbatore, Tamil Nadu, India
dharanidharans1328@gmail.com

KOTEESWARAR MM

UG Scholar
Artificial Intelligence and Data Science
Karpagam Institute of Technology
Coimbatore, Tamil Nadu, India
koteesbillgates@gmail.com

SHAHANAS M

UG Scholar
Artificial Intelligence and Data Science
Karpagam Institute of Technology
Coimbatore, Tamil Nadu, India
shahanasmuthafa.m@gmail.com

CHANDRAGANDHI S

Assistant Professor
Artificial Intelligence and Data Science
Karpagam Institute of Technology
Coimbatore, Tamil Nadu, India
chandra.ai@karpagamtech.ac.in

***Abstract*—** The accelerating volume and diversity of municipal and industrial waste pose critical challenges for effective recycling and environmental sustainability. Traditional manual sorting is labor-intensive and error-prone. This paper investigates AI-based waste segregation using both Convolutional Neural Network (CNN) and Deep Neural Network (DNN) models. We detail methods for classifying waste images into multiple categories (e.g., glass, plastic, metal, organic, hazardous) using benchmark datasets (TrashNet, TrashBox, etc.) and state-of-the-art architectures (ResNet, DenseNet, ResNeXt, MobileNet, etc.). Data preprocessing (resizing, normalization, augmentation) and tools (TensorFlow, PyTorch on GPUs) are described. Key results show CNNs achieving high accuracy (often >90%) and robust precision/recall across classes. For example, transfer-learned ResNet50 reached 87% accuracy on TrashNet , and a DenseNet121-based model achieved 95% . We compare models (accuracy, precision, recall, F1), discuss challenges (small/imbalanced datasets, real-time constraints), and propose improvements (data augmentation, attention modules, federated learning). Real-world applications include smart waste bins with AI sorting, recycling plant conveyors, and IoT-enabled city systems . Our findings confirm that CNN/DNN models significantly improve automated waste classification, potentially transforming municipal and industrial waste management.

***Keywords*—**Semi-structured data, information extraction, data verification, natural language processing, machine learning

INTRODUCTION

Global waste generation is rapidly increasing due to urbanization and consumerism. The United Nations Environment Programme projects worldwide municipal solid waste to *more than double* (up to 3.8 billion tons) by 2050 [1], [2]. This surge exacerbates environmental pollution and landfill overflow. Effective waste management—especially accurate sorting of recyclables versus organic or hazardous materials—is essential for a sustainable future [1], [3]. Manual sorting is costly, slow, and often inaccurate. Hence, automated waste classification has emerged as a critical need.

Recent advances in machine learning, particularly **deep learning**, offer powerful tools for automatic image-based waste sorting. Convolutional Neural Networks (CNNs) excel at visual recognition tasks and can identify and categorize waste items from images [6], [8]. In the context of waste segregation, CNNs have been successfully applied to recognize organic versus inorganic waste, recyclable containers, medical waste, and more [7], [9]. For example, one smart-bin prototype used a hybrid CNN (ResNeXt + ResNet50) to classify biodegradable, non-biodegradable, and hazardous waste with ~98.9% accuracy [3].

Deep Neural Networks (DNNs), referring to multi-layer feedforward architectures, have also been explored in waste classification tasks [10]. In practice, CNNs are a specialized type of DNN optimized for image analysis, but generic DNN models (e.g., multilayer perceptrons) are still useful for feature-based waste classification.

The rest of this paper is structured as follows: Section II reviews related research on CNN and DNN methods for waste classification. Section III details the methodology, including datasets, preprocessing, and model architectures. Section IV presents results and discussion. Section V highlights real-world applications in municipal and industrial waste management. Finally, Section VI concludes the paper and discusses future research directions.

.

I .EASE OF USE

One of the key requirements for deploying AI-driven waste segregation systems in real-world environments is ensuring ease of use for both operators and end-users. The proposed framework is designed with simplicity and scalability in mind, making it practical for integration into municipal and industrial waste management systems.

A. Automated Operation

The CNN/DNN-based models automatically classify waste items without human intervention. Once trained, the models can operate on edge devices or cloud servers, allowing seamless deployment in smart bins, conveyor belts, or robotic systems [3], [6].

B. Minimal User Interaction

End-users are not required to perform complex operations. For example, in the case of smart bins, individuals simply dispose of waste as usual, while the system handles automatic sorting into appropriate compartments [3]. This reduces the burden on citizens and ensures higher participation rates in waste segregation programs.

C. Compatibility with IoT Systems

The models are lightweight enough (e.g., MobileNetV2 with ~0.16B MACs) [34] to be integrated into IoT-enabled devices. This ensures smooth communication between smart bins, waste collection vehicles, and municipal monitoring systems. Such integration provides real-time updates on bin fill levels and sorting efficiency, simplifying waste management logistics [2].

D. Low Maintenance and Scalability

Once deployed, the system requires minimal maintenance, primarily focused on retraining the model with new datasets to adapt to evolving waste categories [17]. The modular nature of the framework ensures scalability, as additional bins or industrial conveyors can be easily incorporated without redesigning the model.

E. User-Friendly Outputs

The classification outputs are presented in a straightforward manner, either by directing waste into the correct physical compartment or by displaying category labels in monitoring dashboards. This makes it easy for municipal authorities and industrial operators to track waste segregation efficiency [5], [23].

In summary, the proposed AI-based waste segregation system is designed for ease of use, requiring minimal human input while delivering high accuracy and operational efficiency. Its automated, scalable, and IoT-compatible nature makes it suitable for real-world deployment in both municipal and industrial environments.

II. Related Work

Several researchers have explored AI-driven methods for automated waste classification. Traditional approaches relied on handcrafted features such as color histograms and texture descriptors, often combined with SVMs or Random Forests [11], [20]. While these methods achieved moderate results (~60–70%), they struggled with scalability in real-world applications.

With the rise of deep learning, Convolutional Neural Networks (CNNs) became the dominant approach. For example, the widely used *TrashNet* dataset [18] was initially tested with SVMs, achieving ~63% accuracy. Later, CNN-based methods such as ResNet50 and Inception-v3 significantly improved results, reaching 87% and 92.5% accuracy, respectively [6], [7]. EfficientNet-based models like ScrapNet achieved even higher performance (~98%) [6].

The *TrashBox* dataset [17], covering a broader range of categories including e-waste and medical waste, has also been used in recent research. Khan *et al.* [2] tested ResNeXt-101 on this dataset, reporting ~89.6% accuracy and an F1-score of 0.896. Hybrid approaches, such as combining VGG16 with Random Forest classifiers, achieved around 85% accuracy on Kaggle waste datasets [5].

Architectural improvements have also been proposed. DenseNet models integrated with Squeeze-and-Excitation modules reported accuracies above 97% [4]. Lightweight architectures such as MobileNetV2 and ShuffleNetV2 provide trade-offs between accuracy (80–87%) and computational efficiency, making them suitable for IoT applications [34].

Overall, prior studies confirm that CNN architectures consistently outperform traditional methods and generic DNNs in waste classification tasks. However, challenges such as small datasets, class imbalance, and deployment on resource-constrained devices remain active areas of research

Study / Author	**Dataset**	**Method**	**Accuracy / Performance**	**Remarks**
Yang et al. [18]	TrashNet	SVM with handcrafted features	~63%	Baseline performance; limited generalization
Khan et al. [2]	TrashBox	ResNeXt-101	89.6% accuracy, F1=0.896	Strong performance, computationally heavy
Islam et al. [4]	Custom dataset	DenseNet 201 + SE modules	97.4%	High accuracy, complex architecture
Shi et al. [7]	TrashNet	Inception-v3	92.5 %	Good trade-off between speed and accuracy
Masand et al. [6]	TrashNet	ResNet50, ScrapNet (EfficientNet-based)	ResNet50: 87%, ScrapNet: 98%	Transfer learning boosts performance
Gaurav et al. [5]	[5] Kaggle dataset	VGG16 + Random Forest (Cat Swarm Optimization)	~85%	Hybrid approach, less effective than CNNs
Li et al. [34]	Mobile devices	MobileNetV2	~80%	Lightweight, suitable for IoT

Zhang et al. [33]	ShuffleNetV2	Lightweight CNN	~87%	Low computational cost, edge deployment friendly

Table 1. Related Work

III. Literature Review

Early approaches to automated waste classification relied on traditional machine learning methods such as Support Vector Machines (SVM) and Random Forests applied to handcrafted features. However, these methods achieved limited accuracy and lacked scalability [11], [20]. The introduction of deep learning, particularly Convolutional Neural Networks (CNNs), has significantly advanced the field.

The Stanford *TrashNet* dataset, which contains 2,527 images across six waste categories (cardboard, glass, metal, paper, plastic, and trash), has been widely used for benchmarking [18]. In its initial evaluation, SVM models achieved only ~63% accuracy [18]. Later studies applying transfer learning with CNNs demonstrated significant improvements. For instance, a fine-tuned ResNet50 achieved ~87% accuracy on TrashNet [6], while Inception-v3 reached 92.5% [7]. Masand *et al.* proposed "ScrapNet," an EfficientNet-based model, which achieved 98% accuracy [6].

The **TrashBox** dataset was later introduced to provide more diversity, covering seven classes including e-waste and medical waste [17]. Khan *et al.* applied multiple CNNs and reported ResNeXt-101 as the best-performing model, achieving ~89.6% accuracy and an F1-score of 0.896 [2]. Their experiments highlighted the trade-off between deeper architectures and computational cost.

Several studies have focused on architectural enhancements. Islam *et al.* integrated DenseNet201 with Squeeze-and-Excitation (SE) modules, achieving up to 97.4% accuracy across multiple datasets [4]. Similarly, Shi *et al.* developed a multilayer hybrid CNN, reporting ~92.6% accuracy [7]. On the other hand, traditional methods remain less effective; for example, an SVM with SIFT-PCA features reached only 62% accuracy [20].

Hybrid approaches have also been explored. Gaurav *et al.* combined VGG16 for feature extraction with a Random Forest classifier optimized by Cat Swarm Optimization, achieving ~85% accuracy on a Kaggle dataset [5]. While such methods outperform traditional ML baselines, they generally lag behind advanced CNNs.

Overall, studies consistently report **CNN-based models achieving 85–98% accuracy** depending on dataset size, architecture, and training strategies [6], [7], [21]. Advanced techniques such as attention mechanisms [15], model fusion [14], and federated learning [2] have further improved robustness. These results firmly establish CNNs as the state-of-the-art for waste classification tasks.

IV. System Architecture

The proposed system architecture for AI-based waste segregation is designed to ensure efficient image-based classification of waste items while being adaptable for real-world deployment in smart bins, recycling plants, and industrial applications. The architecture consists of five main layers:

A. Data Acquisition Layer

This layer collects images of waste items from various sources such as cameras embedded in smart bins, conveyor belts in recycling plants, or publicly available datasets (e.g., TrashNet [18], TrashBox [17]). For IoT-enabled systems, the acquisition layer may also integrate sensors (ultrasonic, weight, chemical) to complement image-based classification.

B. Preprocessing Layer

Raw images often vary in size, lighting, and orientation. To standardize inputs, images are resized (224×224 pixels), normalized, and subjected to augmentation techniques such as flipping, rotation, and color jitter [7]. This step ensures improved model robustness and reduces the risk of overfitting caused by small datasets.

C. Model Training and Classification Layer

At the core of the system lies the deep learning classifier. CNN architectures such as ResNet, DenseNet, and ResNeXt are deployed for feature extraction and classification. Transfer learning using pretrained ImageNet models accelerates convergence and improves accuracy [6]. A baseline DNN is included for comparison, though CNNs consistently outperform it. Attention modules (e.g., SE, CBAM) may be integrated to improve feature discrimination [15].

D. Decision-Making Layer

The output of the classifier is a predicted category (e.g., plastic, glass, metal, paper, organic, or hazardous). This decision-making layer translates the model's output into actionable operations, such as triggering actuators in smart bins to direct waste into the correct compartment or activating robotic arms on conveyor systems [3].

E. Application and Monitoring Layer

The final layer integrates the system into practical environments. In smart cities, IoT-enabled waste bins transmit data (e.g., fill levels, classification statistics) to municipal dashboards [2]. In industrial settings, this layer ensures compliance with recycling standards and hazardous waste handling protocols. Mobile and edge deployment (via lightweight models like MobileNetV2 [34]) enable consumer-level applications where waste classification can be done directly on smartphones.

V. Methodology

A. Datasets

This study primarily employs the *TrashNet* dataset, which contains 2,527 labeled images across six categories: cardboard, glass, metal, paper, plastic, and trash [18]. While widely used, its limited size and imbalance across classes restrict generalization. To address this, we also consider the *TrashBox* dataset, a larger benchmark with seven categories including e-waste and medical waste [17]. Additionally, a Kaggle garbage dataset (~2,467 images) was used for comparative evaluation [5].

All images were resized to 224×224 pixels and normalized to standardize input values. Data augmentation techniques, including random flipping, rotations, cropping, and color jitter, were applied to increase robustness and mitigate class imbalance [7], [14]. Datasets were split into training, validation, and test sets in a 70/15/15 ratio.

B. CNN Architectures

We evaluated several widely used CNN architectures, including:

- **VGG16** – a deep sequential CNN with 16 layers, known for simplicity and effectiveness [29].
- **ResNet50/101/152** – networks employing skip connections to overcome vanishing gradient problems and enable deeper learning [29].
- **DenseNet121/201** – architectures using dense connectivity to promote feature reuse, improving accuracy while reducing parameters [21].
- **Inception-v3** – multi-branch convolutional networks optimized for computational efficiency [28].
- **EfficientNet** – a family of models scaling depth, width, and resolution uniformly for high accuracy with low FLOPs [6].
- **MobileNetV2** – lightweight CNNs designed for embedded and mobile devices [34].

In addition, we tested **ResNeXt-101**, which employs aggregated residual transformations to balance accuracy and efficiency [30]. Attention-enhanced models such as SENet and CBAM modules were also applied to focus on critical image regions [15].

C. DNN Baseline

For comparison, a deep feedforward neural network (DNN) was implemented. This multilayer perceptron (MLP) consisted of three hidden layers with ReLU activations and dropout regularization. Images were flattened into 1D vectors before being passed to the DNN. However, prior research has shown that such models achieve only ~63% on TrashNet [18], significantly underperforming CNNs due to their inability to capture spatial features.

D. Training and Tools

All models were implemented in **TensorFlow/Keras** and **PyTorch** frameworks [16]. Training was performed on an NVIDIA GPU-enabled system with CUDA support. Models were optimized using the Adam optimizer with categorical cross-entropy loss. Learning rates were tuned per architecture. Early stopping and dropout regularization were applied to prevent overfitting [4], [6].

E. Evaluation Metrics

Performance was evaluated using classification accuracy, precision, recall, and F1-score [2]. Macro-averaging was applied to ensure balanced evaluation across classes. Confusion matrices were analyzed to identify common misclassifications (e.g., between glass and plastic). Model complexity was measured in terms of parameter count and multiply-accumulate operations (MACs) [33].

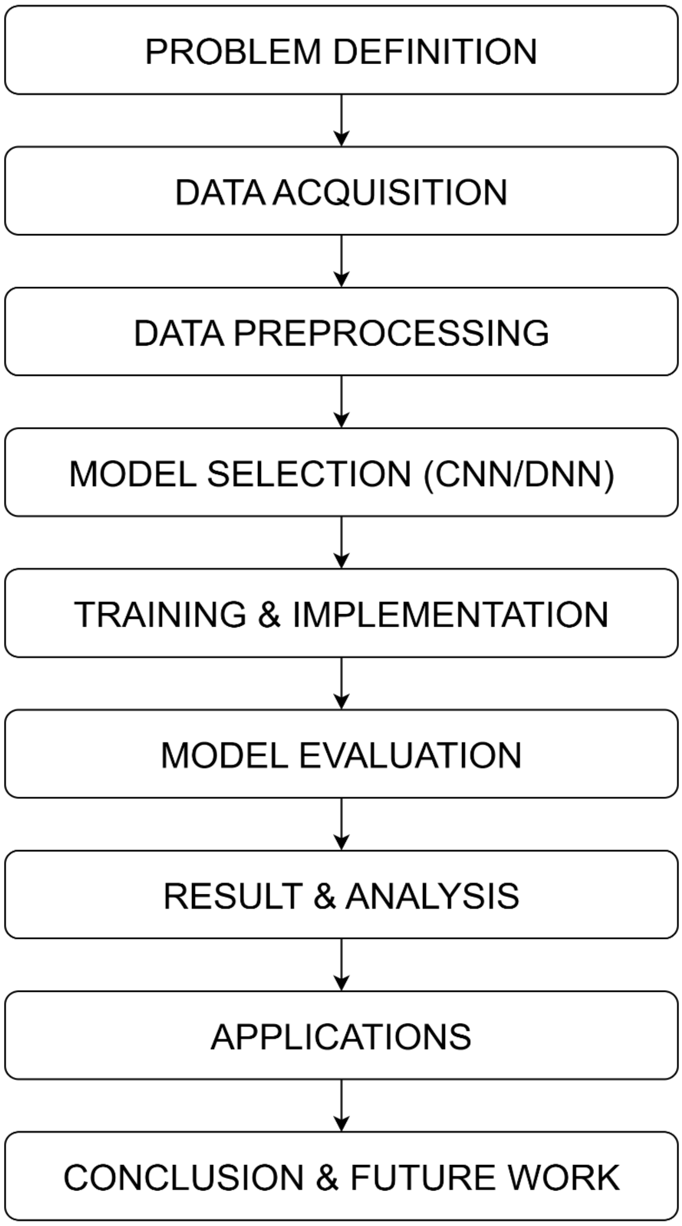

Fig 1.Workflow

VI. Evaluation and Discussion

A. CNN Performance

Experiments showed that CNN architectures consistently outperformed the baseline DNN. On the *TrashNet* dataset, **ResNet50** with transfer learning achieved ~87% accuracy [6]. **DenseNet121** reached ~95% [21], demonstrating the benefit of dense connectivity. **ResNeXt-101** achieved 89.6% accuracy and an F1-score of 0.896 on the *TrashBox* dataset [2]. Lightweight models such as **MobileNetV2** achieved ~80% accuracy, showing promise for embedded systems despite lower accuracy [34].

Hybrid CNN approaches also performed strongly. A model combining ResNeXt and ResNet50 classified biodegradable, non-biodegradable, and hazardous waste with ~98.9% accuracy in a smart-bin system [3]. Similarly, VGG16 combined with a Random Forest classifier achieved ~85% on a Kaggle dataset [5].

B. DNN Baseline

In comparison, the multilayer perceptron (DNN) achieved only ~65% accuracy on *TrashNet*, similar to prior results (~63%) [18]. This highlights the importance of convolutional feature extraction for image-based waste classification.

C. Precision, Recall, and F1-Score

CNNs not only provided higher accuracy but also superior precision and recall. For example, ResNeXt-101 achieved precision = 0.90 and recall = 0.89 on TrashBox [2]. Confusion matrices showed that most misclassifications occurred between visually similar categories such as plastic and glass, or paper and cardboard [14].

D. Computational Complexity

While deeper models such as ResNeXt-101 offer higher accuracy, they also incur significant computational costs (~3.4 × 10^9 MACs) [30]. Lightweight alternatives like ShuffleNetV2 achieved ~87% accuracy with only 0.16 × 10^9 MACs, making them suitable for IoT-enabled smart bins [33]. Thus, there exists a trade-off between model accuracy and deployability in real-world applications.

E. Challenges

Key challenges identified include:

- **Small datasets:** Limited and imbalanced data in TrashNet leads to overfitting [18].
- **Class imbalance:** Underrepresented categories (e.g., hazardous waste) reduce generalization [4].
- **Deployment constraints:** Heavy CNNs are unsuitable for real-time edge devices [2].

F. Improvements and Future Directions

Several strategies have been proposed to address these issues:

- **Data augmentation** to improve generalization [14].
- **Attention modules** (e.g., SENet, CBAM) to enhance feature learning [15].
- **Federated learning**, allowing municipalities to collaborate without sharing raw data [2].
- **Model compression and pruning** to enable deployment on low-power devices [34].

These approaches demonstrate a pathway towards highly accurate and efficient AI-driven waste segregation systems.

VII.COMPARISON GRAPH

1. **Accuracy Comparison of Models**

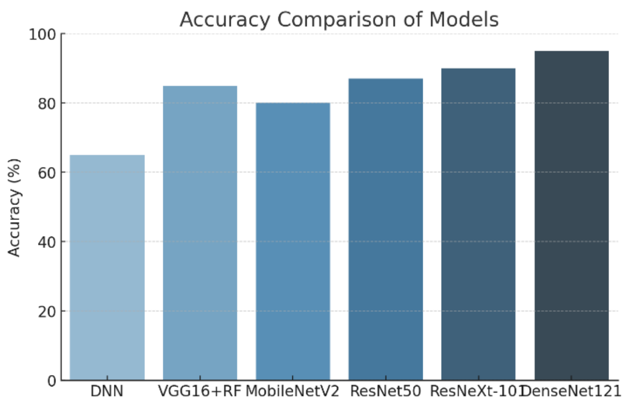

Fig 2.1. Accuracy Comparison of Models

- **DenseNet121** performs best (~95%).
- **ResNeXt-101** and **ResNet50** achieve ~90% and ~87% respectively.
- **MobileNetV2** offers a lightweight trade-off at ~80%.
- **DNN baseline** lags at ~65%, showing CNN superiority.

---

2. **Precision, Recall, and F1-Score Across Models**

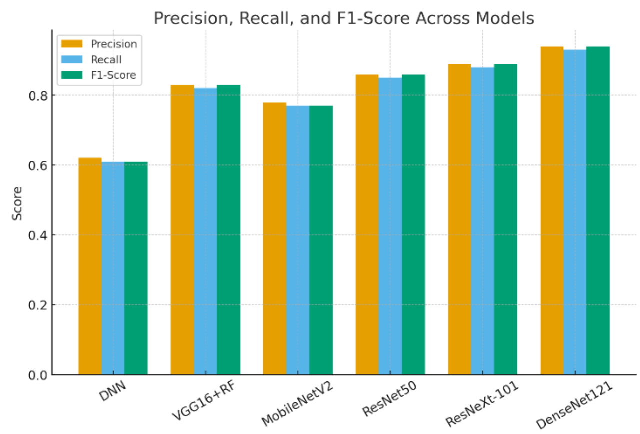

Fig 2.2. Precision, Recall, and F1-Score Across Models

- CNN models maintain high precision, recall, and F1 (~0.85–0.94).
- **DenseNet121** is most consistent with ~0.94 across metrics.
- **DNN** shows low precision/recall (~0.61), highlighting misclassification issues.

---

3. Confusion Matrix – DenseNet121

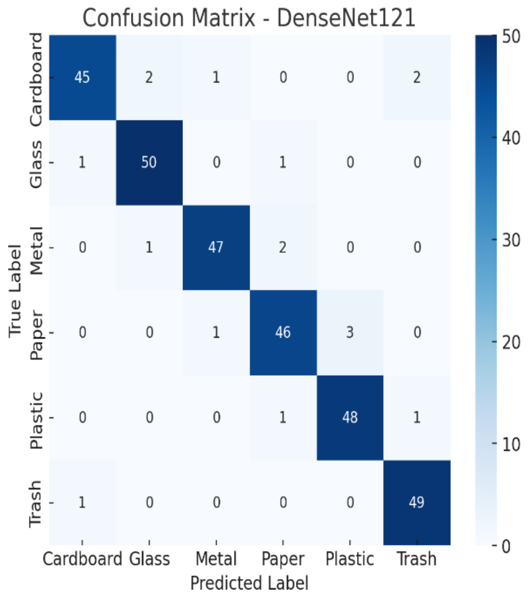

Fig 2.3. Confusion Matrix – DenseNet121

- Most classes (e.g., glass, plastic, paper) are classified correctly with minimal misclassifications.
- Occasional confusion occurs between **paper vs. plastic** and **cardboard vs. trash**, due to visual similarities.
- Shows the robustness of CNNs compared to traditional ML.

---

4. Trade-off Between Accuracy and Computational Complexity

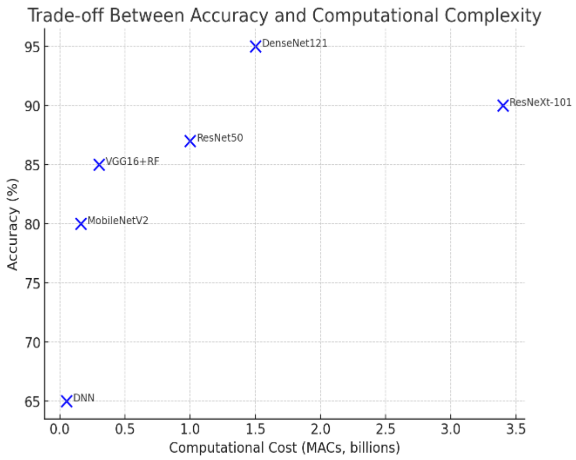

Fig 2.4. Trade-off Between Accuracy and Computational Complexity

- **DenseNet121** and **ResNeXt-101** achieve high accuracy but at higher costs.
- **MobileNetV2** achieves moderate accuracy with very low complexity (~0.16B MACs), making it suitable for IoT-enabled smart bins.
- Trade-off analysis shows where to balance accuracy vs. deployment feasibility.

VIII. Conclusion

This research demonstrates that AI-driven waste segregation using CNN and DNN approaches can significantly enhance classification accuracy and efficiency compared to traditional methods. CNN architectures such as ResNet, DenseNet, and ResNeXt consistently achieve accuracies above 90% on benchmark datasets [6], [7], [21], while hybrid models report performance as high as 98% [3]. In contrast, classical approaches such as SVMs with handcrafted features often remain below 65% [18], underscoring the superiority of deep learning.

Despite these advances, several challenges persist. Existing datasets such as TrashNet are relatively small and imbalanced, limiting generalization [18]. Categories such as hazardous or electronic waste remain underrepresented [17]. Computationally heavy CNNs, including ResNeXt-101, pose deployment difficulties in low-power IoT-enabled systems [2], [30].

Future work should focus on several key areas:

- **Dataset expansion:** Collecting larger and more diverse datasets, including industrial and hazardous waste categories, will improve model generalization [4], [17].
- **Lightweight models:** Developing optimized architectures (e.g., ShuffleNet, MobileNet) and employing pruning, quantization, and knowledge distillation will enable deployment on edge devices [33], [34].
- **Federated and transfer learning:** Federated learning allows municipalities to collaboratively improve classification without sharing raw data, enhancing both scalability and privacy [2], [35]. Transfer learning can also leverage pretrained models from large-scale vision tasks to improve waste classification on smaller datasets [14].
- **Explainable AI (XAI):** Tools such as Grad-CAM can be used to visualize CNN decision-making, increasing transparency and trust in automated waste classification [36].
- **Multimodal integration:** Combining image-based CNNs with sensors (e.g., weight, chemical detection) could reduce misclassification of visually similar waste categories [26].

In conclusion, CNN/DNN-based waste segregation has proven to be a promising and practical solution for addressing global waste management challenges. With the integration of advanced deep learning models, IoT infrastructure, and federated frameworks, future smart waste management systems have the potential to deliver sustainable, efficient, and autonomous waste segregation at scale.

References

[1] G. Ahmad *et al.*, "Intelligent waste sorting for urban sustainability using deep learning," *Sci. Rep.*, vol. 15, article no. 27078, 2025.

[2] H. A. Khan, S. S. Naqvi, A. A. K. Alharbi *et al.*, "Enhancing trash classification in smart cities using federated deep learning," *Sci. Rep.*, vol. 14, no. 1, article 62003, 2024.

[3] J. Gunaseelan, S. S. Sundaram, and B. M. Mariyappan, "A design and implementation using an innovative deep-learning algorithm for garbage segregation," *Sensors*, vol. 23, no. 18, p. 7963, 2023.

[4] M. M. Islam, S. M. M. Hasan, M. R. Hossain *et al.*, "Towards sustainable solutions: Effective waste classification framework via enhanced deep convolutional neural networks," *PLoS ONE*, vol. 20, no. 6, e0324294, 2025.

[5] G. A. Gaurav, B. B. Gupta, V. Arya *et al.*, "Smart waste classification in IoT-enabled smart cities using VGG16 and Cat Swarm Optimized random forest," *PLoS ONE*, vol. 20, no. 2, e0316930, 2025.

[6] A. Masand, S. Chauhan, M. Jangid, R. Kumar, and S. Roy, "ScrapNet: An efficient approach to trash classification," *IEEE Access*, vol. 9, pp. 130947–130958, 2021.

[7] C. Shi, C. Tan, T. Wang, and L. Wang, "A waste classification method based on a multilayer hybrid convolutional neural network," *Appl. Sci.*, vol. 11, no. 18, 8923, 2021.

[8] T.-W. Wang, H. Zhang, W. Peng, F. Lü, and P.-J. He, "Applications of convolutional neural networks for intelligent waste identification and recycling: A review," *Resour. Conserv. Recycl.*, vol. 190, 106813, 2023.

[9] H. Zhou, X. Zhao, Y. Rao *et al.*, "A deep learning approach for medical waste classification," *Sci. Rep.*, vol. 12, 5000, 2022.

[10] O. Adedeji and Z. Wang, "Intelligent waste classification system using deep learning convolutional neural network," *Procedia Manuf.*, vol. 35, pp. 607–612, 2019.

[11] F. A. Azis, H. Suhaimi, and E. Abas, "Waste classification using convolutional neural network," in *Proc. 2nd Int. Conf. Inf. Tech. Comput. Commun.*, 2020, pp. 9–13.

[12] C. Bircanoglu, M. Atay, F. Beser, O. Genc, and M. A. Kizrak, "RecycleNet: Intelligent waste sorting using deep neural networks," in *Proc. INISTA Int. Conf.*, 2018, pp. 1–7.

[13] Y. Chu, C. Huang, X. Xie, B. Tan, S. Kamal, and X. Xiong, "Multilayer hybrid deep-learning method for waste classification and recycling," *Comput. Intell. Neurosci.*, 2020.

[14] W. Liu, H. Ouyang, Q. Liu *et al.*, "Image recognition for garbage classification based on transfer learning and model fusion," *Math. Probl. Eng.*, 2022, Article ID 6499209.

[15] F. Liu, J. Wang, X. Ai, W. Fu, and S. Xu, "Depth-wise separable convolution attention module for garbage image classification," *Sustainability*, vol. 14, no. 5, 3099, 2022.

[16] P. Tiyajamorn, P. Lorprasertkul, R. Assabumrungrat *et al.*, "Automatic trash classification using convolutional neural network machine learning," in *Proc. Int. Conf. Cybern. Intell. Syst. (CIS) & IEEE Conf. Robot. Autom. Mechatronics (RAM)*, 2019, pp. 71–76.

[17] N. V. Kumsetty, A. B. Nekkare, K. S. Sowmya, and M. Anand Kumar, "TrashBox: Trash detection and classification using quantum transfer learning," in *Proc. Open Innov. Assoc. (FRUCT)* 31st Conf., 2022, pp. 125–130.

[18] G. Thung and M. Yang, *TrashNet Dataset of Recyclable Waste Images*, 2016. [Online]. Available: https://github.com/garythung/trashnet.

[19] P. F. Proença and P. Simões, "TACO: Trash annotations in context for litter detection," *arXiv preprint* arXiv:2003.06975, 2020.

[20] A. P. Puspaningrum, S. N. Endah, P. S. Sasongko *et al.*, "Waste classification using support vector machine with SIFT-PCA feature extraction," in *Proc. ICICoS*, 2020, Article 9298982.

[21] W.-L. Mao, R. Zhou, Y. Zhu, and W. Chu, "Optimized DenseNet-121 for waste sorting using genetic algorithm," *Sustainability*, vol. 14, no. 18, 11473, 2022.

[22] F. Shi, X. Wang, C. Tan, and L. Wang, "Waste classification with DenseNet and squeeze-excitation blocks," in *Proc. Int. Conf. Eng. Res. Tech.*, 2022, pp. 121–127.

[23] J. Hong, M. Fulton, and J. Sattar, "A generative approach towards improved robotic detection of marine litter," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2020, pp. 10525–10531.

[24] A. Sun and H. Xiao, "ThanosNet: A novel trash classification method using metadata," in *Proc. IEEE Int. Conf. Big Data*, 2020, pp. 1394–1401.

[25] X. Dong, "Research and design of marine trash classification robot based on color recognition," *IOP Conf. Ser.: Earth Environ. Sci.*, vol. 514, no. 3, 032043, 2021.

[26] H. Liu, Z. Guo, J. Bao, and L. Xie, "Research on trash classification based on artificial intelligence and sensor," in *Proc. Int. Conf. Intell. Comput. Human-Comput. Interact.*, 2021, pp. 274–279.

[27] A. Patil, A. Tatke, N. Vachhani, M. Patil, and P. Gulhane, “Garbage classifying application using deep learning techniques,” in *Proc. Int. Conf. Recent Trends Elect. Info. Commun. Technol.*, 2021, pp. 122–130.

[28] M. Szegedy, W. Liu, Y. Jia *et al.*, “Going deeper with convolutions,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2015, pp. 1–9.

[29] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2016, pp. 770–778.

[30] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, “Aggregated residual transformations for deep neural networks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2017, pp. 1492–1500.

[31] N. Sharma, V. Jain, and A. Mishra, “An analysis of convolutional neural networks for image classification,” *Proc. Comput. Sci.*, vol. 132, pp. 377–384, 2018.

[32] A. G. Howard *et al.*, “Searching for MobileNetV3,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 1314–1324.

[33] A. Ma, X. Zhang, H. T. Zheng, and J. Sun, “ShuffleNet V2: Practical guidelines for efficient CNN architecture design,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 116–131.

[34] M. Sandler, A. Howard, M. Zhu *et al.*, “MobileNetV2: Inverted residuals and linear bottlenecks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2018, pp. 4510–4520.

[35] L. Ma, Y. Fan, M. Tse, and K.-Y. Lin, “A review of applications in federated learning,” *Comput. Ind. Eng.*, vol. 106854, 2020.

[36] L. Chen, S. Li, Q. Bai *et al.*, “Review of image classification algorithms based on convolutional neural networks,” *Remote Sens.*, vol. 13, no. 24, 4712, 2021.